

4	C	If V_{ice} is volume of ice cube V_{sub} is volume of ice cube submerged in salt solution V_{water} is volume of water obtain when ice cube has melted to water Weight of salt water displaced = Weight of ice cube $\rho_{salt} V_{Sub} = \rho_{ice} V_{ice}$ Mass of ice = Mass of water it melts into $\rho_{ice} V_{ice} = \rho_{Water} V_{water}$ Hence $\rho_{Water} V_{water} = \rho_{salt} V_{Sub}$ $V_{water} = \frac{\rho_{salt}}{\rho_{Water}} V_{Sub}$ And since $\rho_{salt} \geq \rho_{Water}$ $V_{water} \geq V_{Sub}$ And hence water will rise.
5	C	Regardless of the nature of the wall it will exert a normal contact force